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Rethinking the Design Space of Vanilla Pixel Diffusion Transformers

MFlow Learns a Per-Direction Interpolation Between x0- and Velocity Prediction, Closing the Quality Gap Between Pixel and Latent Diffusion

By Soumik Mukhopadhyay, Long Mai, Abhinav Shrivastava, Aniruddha Mahapatra

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Pixel-space diffusion transformers avoid VAE tokenization artifacts but have lagged behind latent diffusion in generation quality, partly because the optimal prediction parameterization—clean image (x0) versus velocity (u)—depends non-trivially on model capacity and data statistics. MFlow addresses this by introducing per-principal-direction gains that interpolate between x0- and u-prediction, learned end-to-end with diffusion training. Experiments on ImageNet, UCF-101 video, and large-scale text-to-image generation confirm that MFlow consistently outperforms fixed-parameterization baselines, and that larger models systematically prefer velocity prediction. A lightweight ConvRefiner trained post-hoc with an adversarial loss further closes the persistent high-frequency gap.

AT A GLANCE

Problem: The best prediction parameterization for pixel DiTs varies with model capacity and data; the fixed-choice approach leaves quality on the table.

Why it matters: Pixel DiTs promise simpler, higher-fidelity generation than latent diffusion, but their quality gap has blocked adoption.

Method: Per-direction gain m_i in [0,1] interpolates x0- and u-prediction; gains trained jointly with the diffusion model. ConvRefiner handles residual high-frequency deficits post-hoc.

Result: Competitive or improved FID/FVD vs. fixed-parameterization baselines on ImageNet, UCF-101, and text-to-image.

Evidence: Ablations across model scales, architectures, and modalities; systematic m_i analysis validating the capacity-preference relationship.

THE BIG PICTURE

Latent diffusion models (LDMs) such as SD3 and FLUX compress images into a low-dimensional latent code before diffusion, enabling fast generation but introducing VAE reconstruction artifacts. Pixel diffusion transformers (pixel DiTs) sidestep this by diffusing directly in pixel space, preserving sharp boundaries.

The quality gap between pixel DiTs and LDMs has been attributed to less stable training. This paper identifies an underappreciated cause: the prediction parameterization. The choice between predicting the clean image (x0), the noise (epsilon), or the velocity (u = alpha*x0 - sigma*epsilon) determines the gradient signal across noise levels in ways not studied systematically for pixel DiTs.

WHY EXISTING APPROACHES FALL SHORT

Fixed x0-prediction (JiT): Showed x0-prediction outperforms epsilon-prediction at moderate scale, but this advantage diminishes or reverses at larger model sizes.

Fixed u-prediction (flow matching): Works well for latent diffusion but underperforms x0-prediction at smaller pixel DiT scales.

AsymFlow: A binary rank-asymmetric assignment is a special case of MFlow with step-function gains, losing the benefit of smooth interpolation.

CORE MECHANISM

MFlow parameterizes the prediction target for the i-th principal direction as:

y_hat_i = m_i * u_i + (1 - m_i) * (x0)_i

where the scalar gain m_i in [0,1] is the sole additional parameter per direction. m_i = 0 recovers clean-image prediction; m_i = 1 recovers velocity prediction. Gains are initialized from signal-to-noise statistics and updated with gradient descent.

ConvRefiner is a lightweight CNN trained to predict the residual in the frequency domain with a spectral adversarial loss, with the diffusion backbone frozen.

TECHNICAL FORMULATION

Flow matching diffusion: x_t = alpha_t * x0 + sigma_t * epsilon.

Velocity target: u = alpha_t * x0 - sigma_t * epsilon.

MFlow model f_theta predicts:

y_hat_i(x_t, t) = m_i * f_theta^(u)(x_t, t)_i + (1-m_i) * f_theta^(x0)(x_t, t)_i

Training loss: L_MFlow = E_{t,x0,eps}[sum_i ||y_hat_i - y_i^target||^2]

Gradients flow through both f_theta and {m_i}.

Optimal gain under linear model: m_i* = Var[u_i] / (Var[u_i] + Var[(x0)_i])

LEARNING OR INFERENCE PROCEDURE

Training: Standard diffusion training with an extra gradient term for {m_i}. Gains initialized as m_i = SNR_i / (1 + SNR_i). ConvRefiner trained in a second stage with backbone frozen.

Inference: Identical to standard flow-matching sampling (DDIM, DPM-Solver, or Euler). ConvRefiner adds one forward pass with the lightweight CNN.

WHAT THE GUARANTEE SAYS

MFlow is a strict generalization of fixed x0- and u-prediction: setting all m_i = 0 recovers the x0 loss; all m_i = 1 recovers velocity loss. Therefore MFlow can only improve on either fixed choice—its optima weakly dominate.

The theoretical optimal m_i* is an increasing function of intrinsic dimensionality, explaining why larger models (which represent more directions) benefit more from velocity prediction.

EXPERIMENTAL FINDINGS

ImageNet 256x256: MFlow reduces FID by 15–20% over fixed x0- and u-prediction baselines at the same model scale.

UCF-101 video: Consistent FVD improvement; gains are largest for the largest model (+18% over velocity-only baseline).

Large-scale text-to-image: MFlow at 1B parameters matches or exceeds prior best pixel DiT at 3B parameters.

ConvRefiner: Adds 1.2 ms per image; reduces high-frequency FID deficit by 40% without changing global FID.

Capacity scaling: Empirical m_i values shift from near-zero to near-one as model size increases, confirming the theoretical prediction.

ABLATIONS AND INTERPRETATION

Gain initialization: Random initialization converges to similar final values but requires 2x more training steps; SNR-based initialization strongly recommended.

Per-direction vs. global gain: A single shared gain m reduces FID by only 60% of the per-direction improvement.

ConvRefiner training signal: Replacing spectral GAN with pixel-space L2 loss fails to recover high-frequency detail—adversarial objective is essential.

REFERENCE

Soumik Mukhopadhyay, Long Mai, Abhinav Shrivastava, Aniruddha Mahapatra. **Rethinking the Design Space of Vanilla Pixel Diffusion Transformers.** arXiv:2610.23854. Under review, 2026. https://arxiv.org/abs/2610.23854